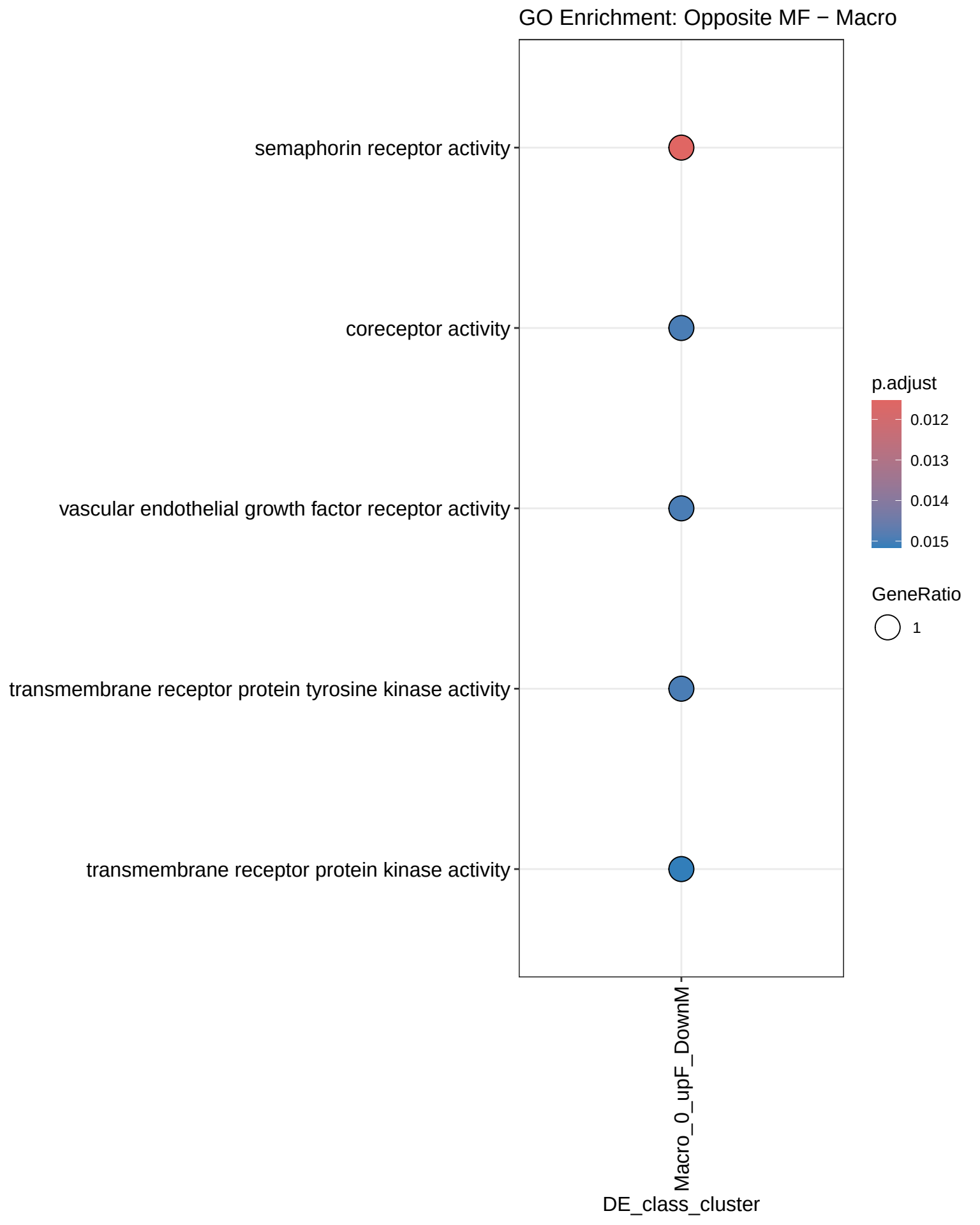
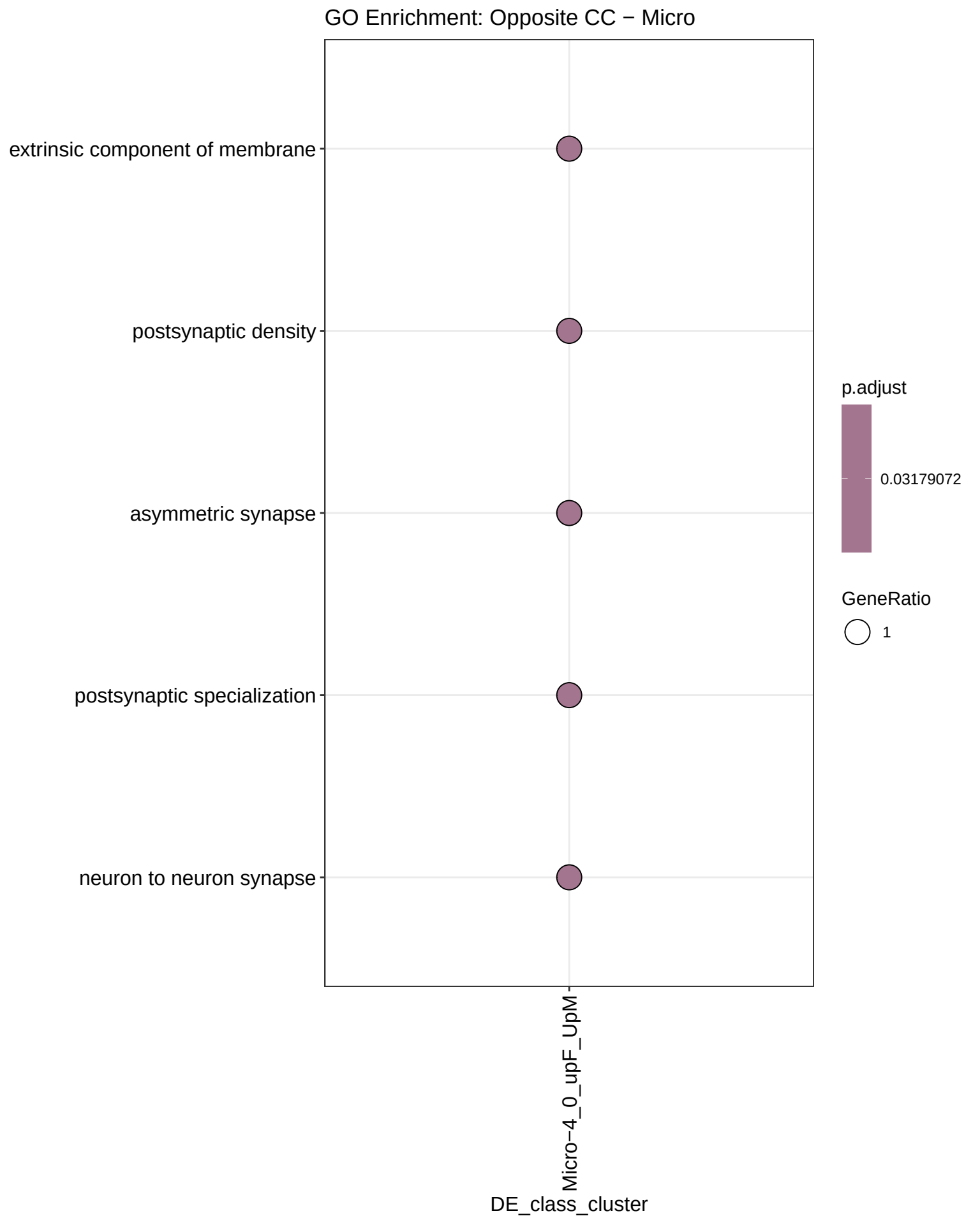
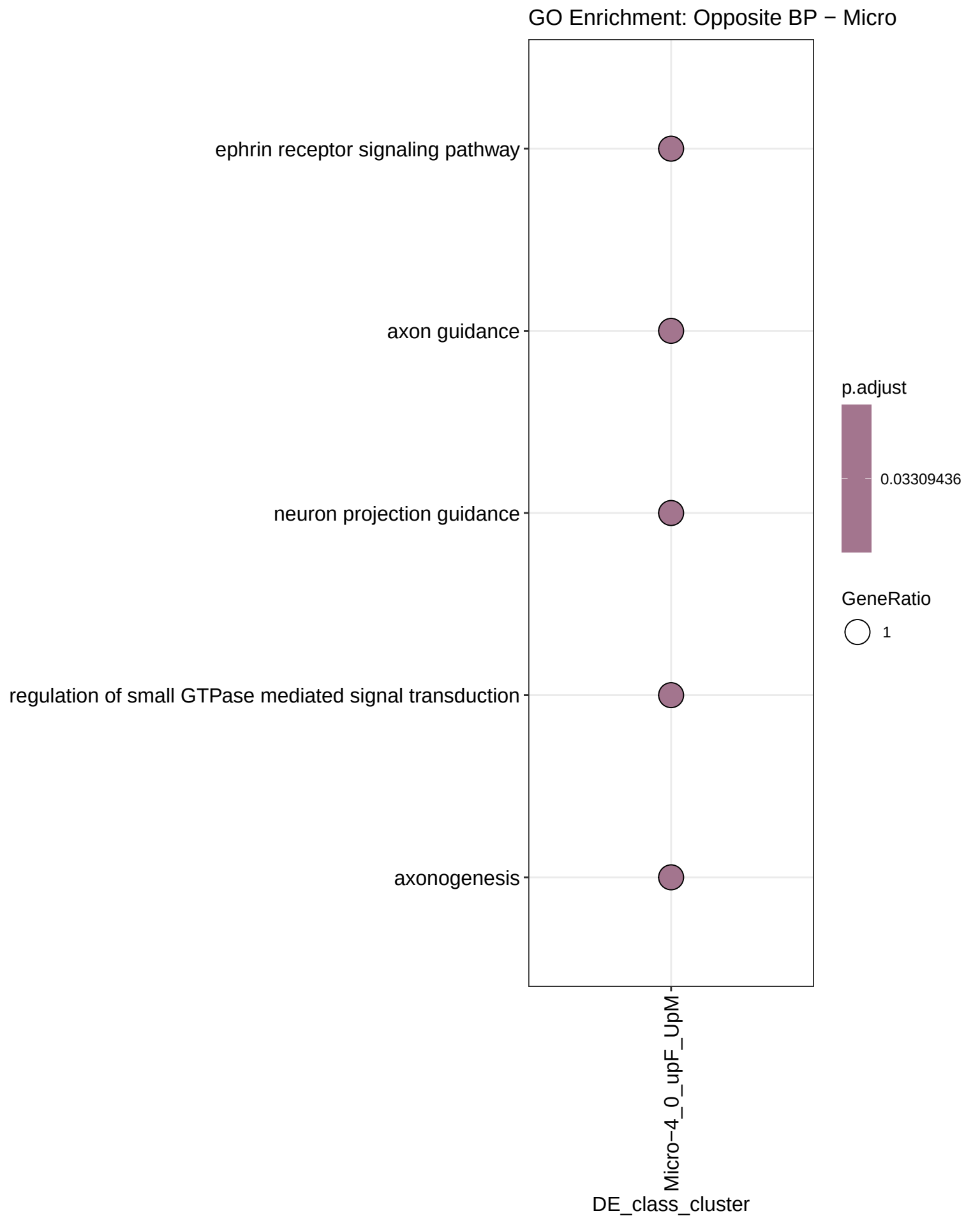
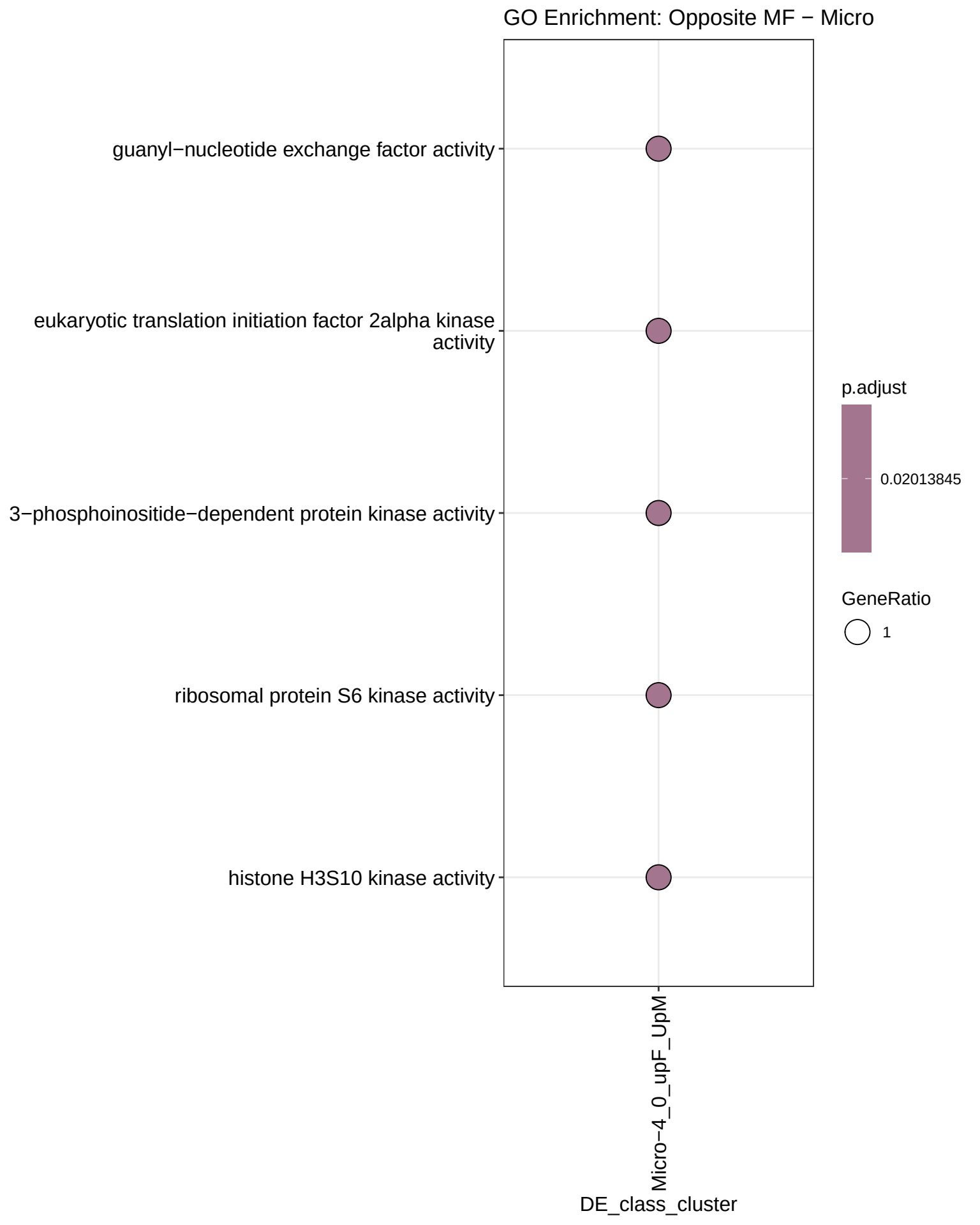
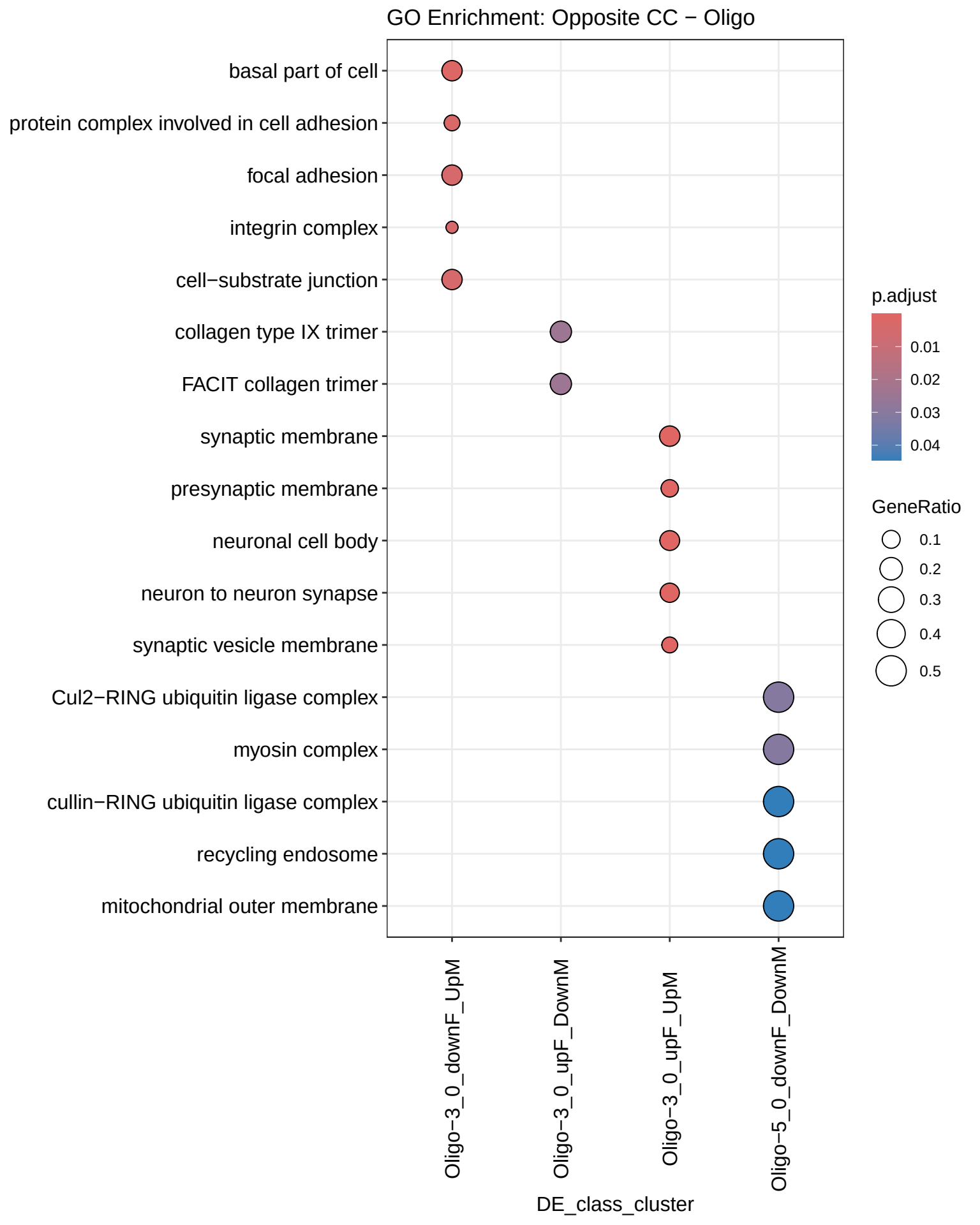
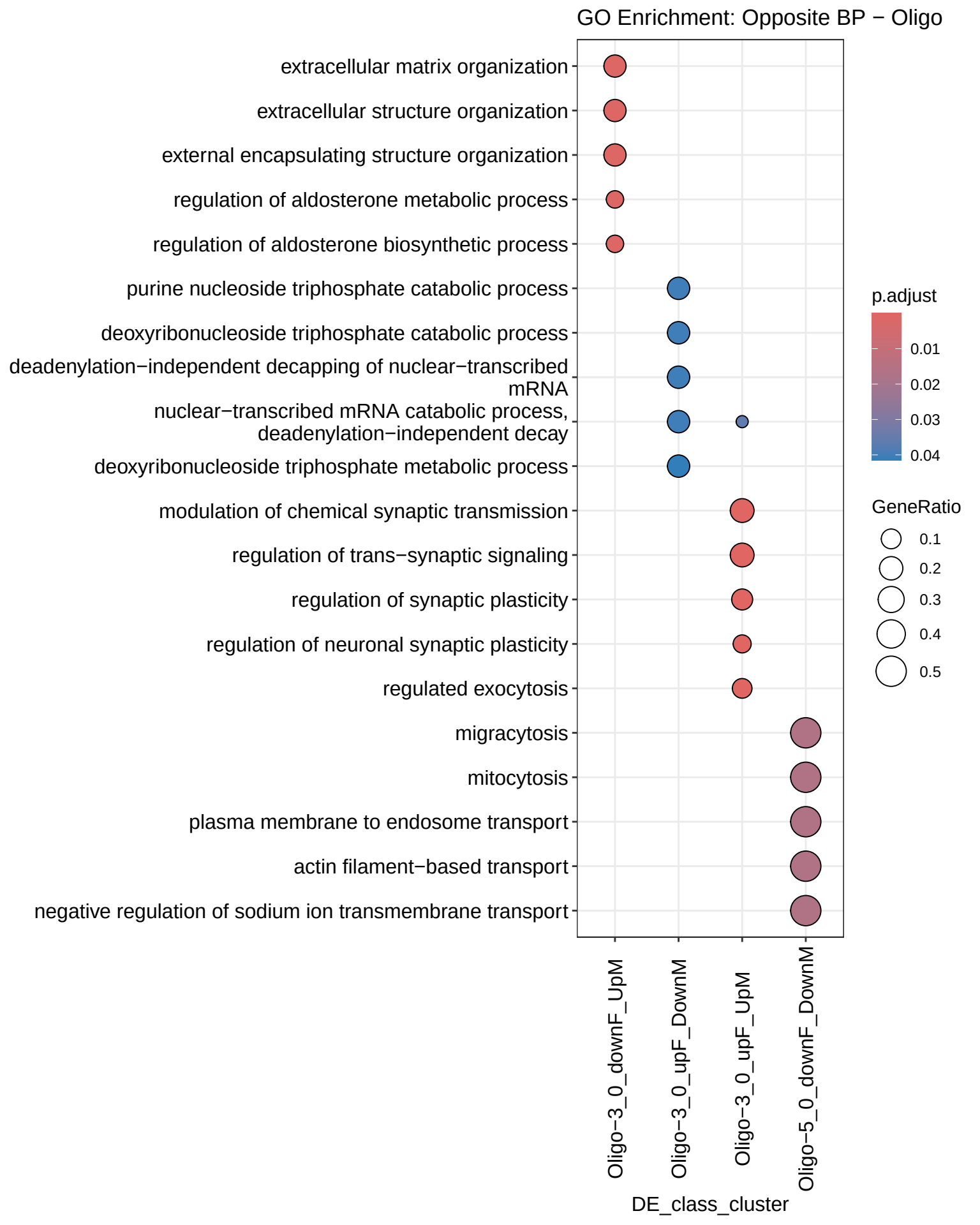

GO Enrichment: Opposite MF − Macro
semaphorin receptor activity
coreceptor activity
vascular endothelial growth factor receptor activity
transmembrane receptor protein tyrosine kinase activity
transmembrane receptor protein kinase activity
Macro_0_upF_DownM
DE_class_cluster
p.adjust
0.012
0.013
0.014
0.015
GeneRatio
1

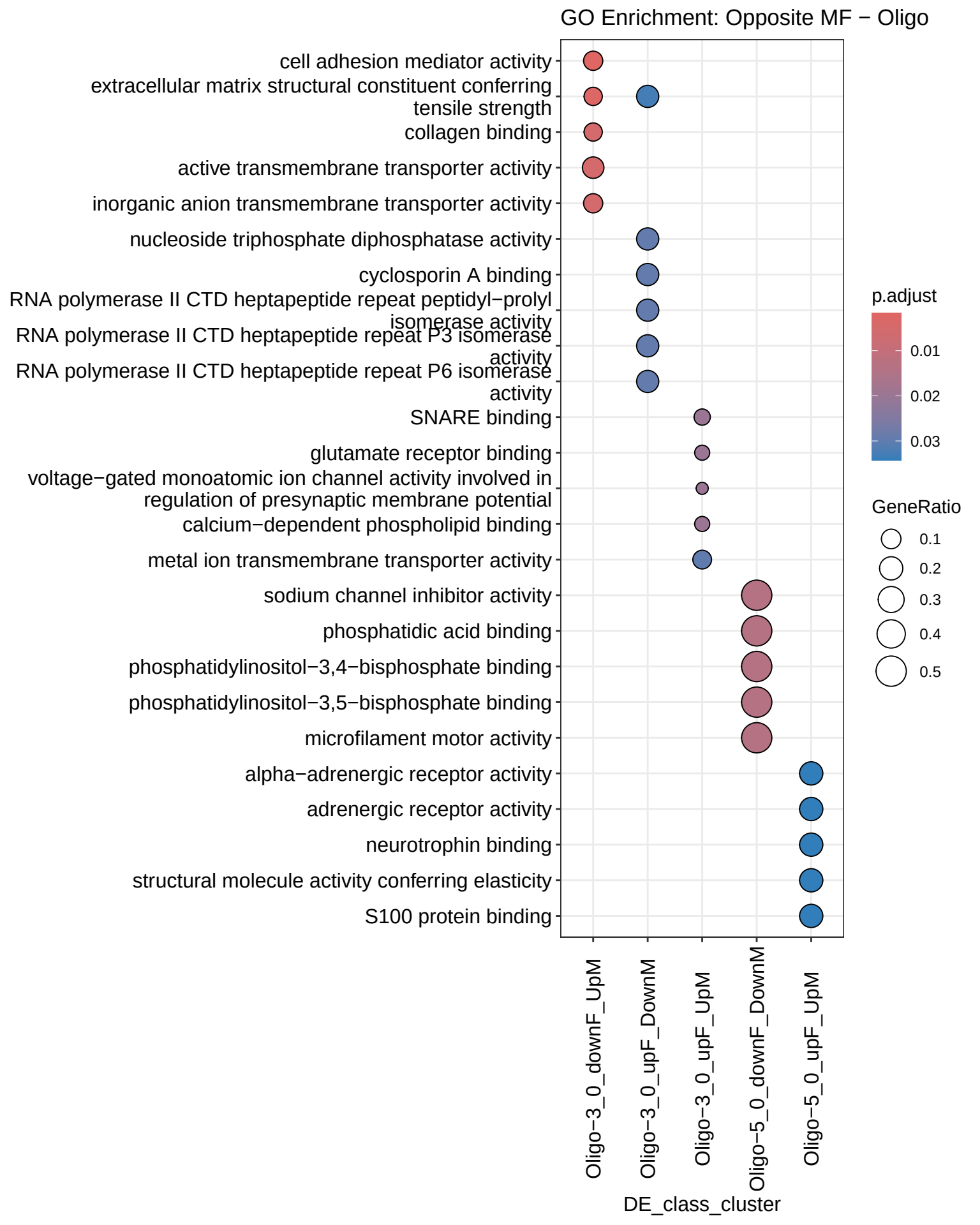

GO Enrichment: Opposite CC − Micro
extrinsic component of membrane
postsynaptic density
asymmetric synapse
postsynaptic specialization
neuron to neuron synapse
p.adjust
0.03179072
GeneRatio
1
Micro−4_0_upF_UpM
DE_class_cluster

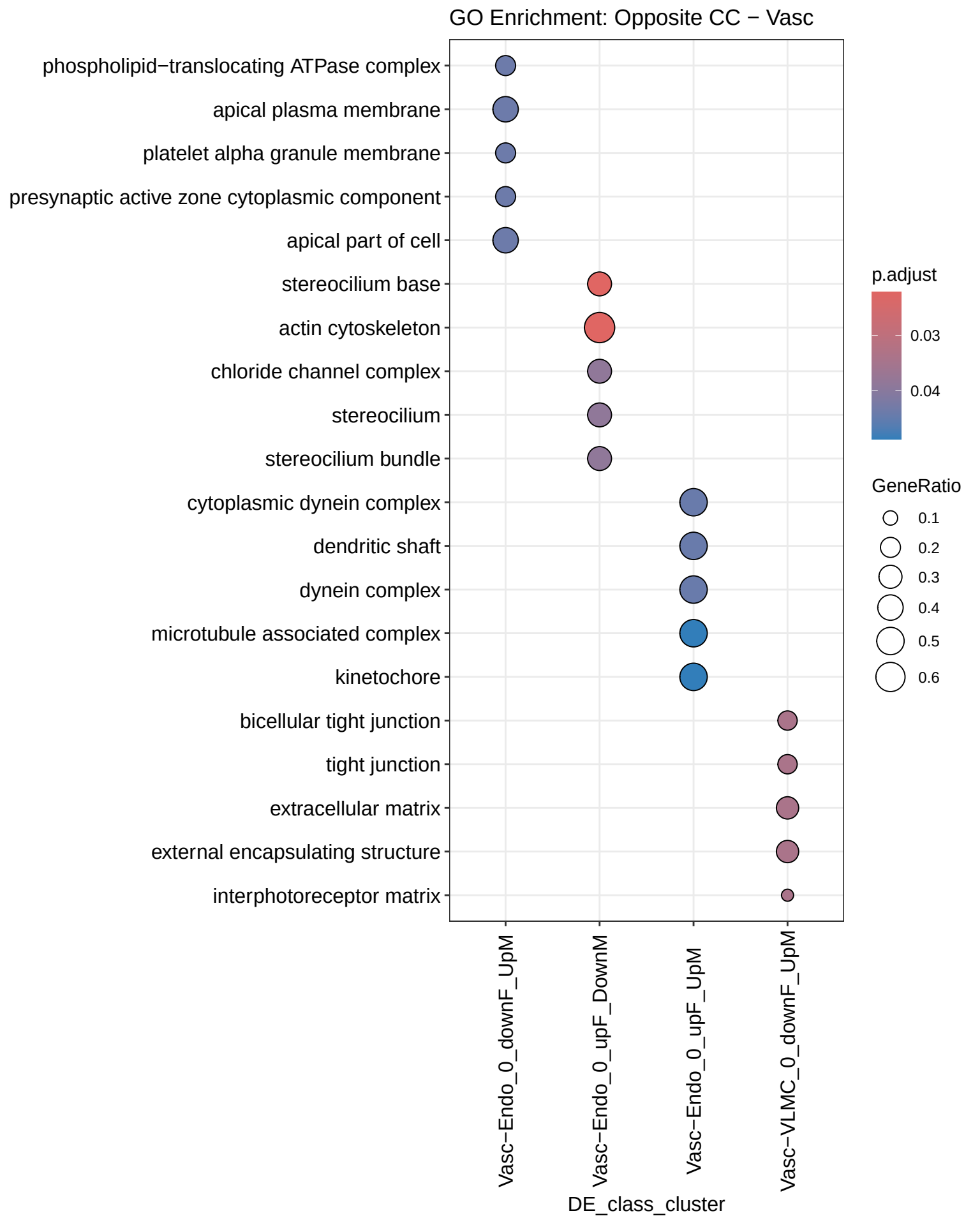

GO Enrichment: Opposite BP − Micro
ephrin receptor signaling pathway
axon guidance
neuron projection guidance
regulation of small GTPase mediated signal transduction
axonogenesis
p.adjust
0.03309436
GeneRatio
1
Micro−4_0_upF_UpM
DE_class_cluster

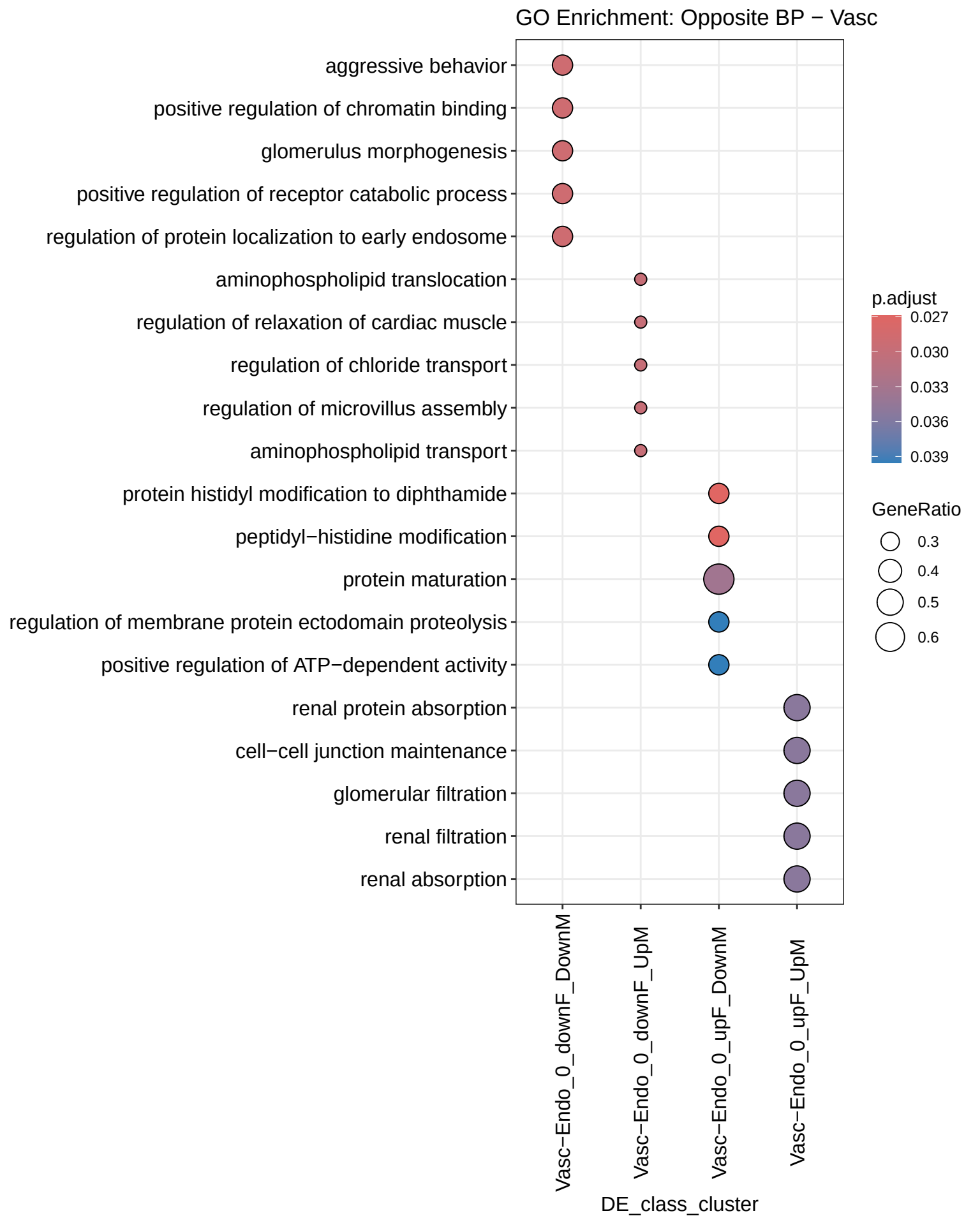

GO Enrichment: Opposite MF − Micro
guanyl−nucleotide exchange factor activity
eukaryotic translation initiation factor 2alpha kinase activity
3−phosphoinositide−dependent protein kinase activity
ribosomal protein S6 kinase activity
histone H3S10 kinase activity
Micro−4_0_upF_UpM
DE_class_cluster
p.adjust
0.02013845
GeneRatio
1


GO Enrichment: Opposite CC − Oligo
basal part of cell
protein complex involved in cell adhesion
focal adhesion
integrin complex
cell−substrate junction
collagen type IX trimer
FACIT collagen trimer
synaptic membrane
presynaptic membrane
neuronal cell body
neuron to neuron synapse
synaptic vesicle membrane
Cul2−RING ubiquitin ligase complex
myosin complex
cullin−RING ubiquitin ligase complex
recycling endosome
mitochondrial outer membrane
Oligo−3_0_downF_UpM
Oligo−3_0_upF_DownM
Oligo−3_0_upF_UpM
Oligo−5_0_downF_DownM
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.1
0.2
0.3
0.4
0.5

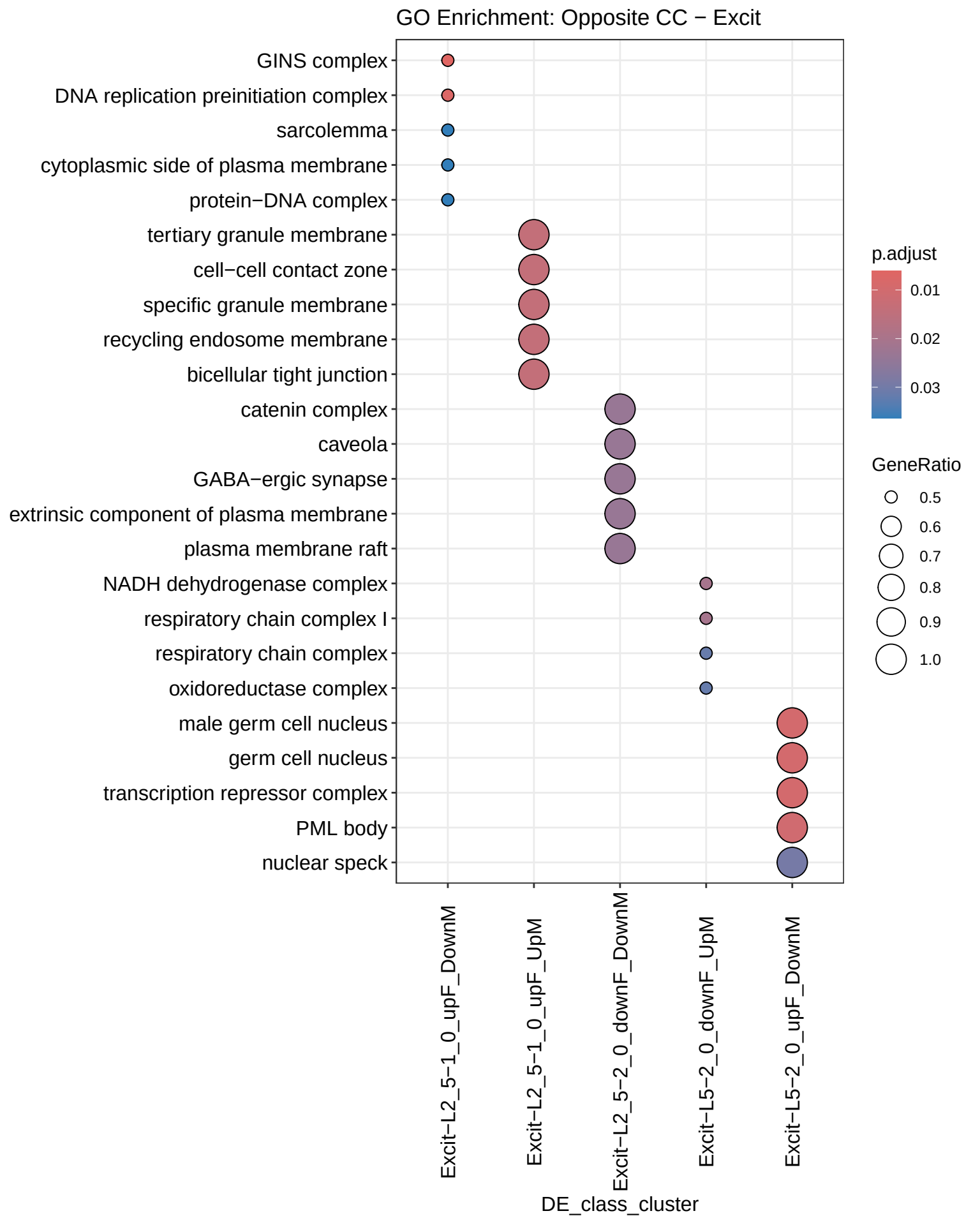

GO Enrichment: Opposite BP − Oligo
extracellular matrix organization
extracellular structure organization
external encapsulating structure organization
regulation of aldosterone metabolic process
regulation of aldosterone biosynthetic process
purine nucleoside triphosphate catabolic process
deoxyribonucleoside triphosphate catabolic process
deadenylation−independent decapping of nuclear−transcribed mRNA
nuclear−transcribed mRNA catabolic process, deadenylation−independent decay
deoxyribonucleoside triphosphate metabolic process
modulation of chemical synaptic transmission
regulation of trans−synaptic signaling
regulation of synaptic plasticity
regulation of neuronal synaptic plasticity
regulated exocytosis
migracytosis
mitocytosis
plasma membrane to endosome transport
actin filament−based transport
negative regulation of sodium ion transmembrane transport
Oligo−3_0_downF_UpM
Oligo−3_0_upF_DownM
Oligo−3_0_upF_UpM
Oligo−5_0_downF_DownM
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.1
0.2
0.3
0.4
0.5

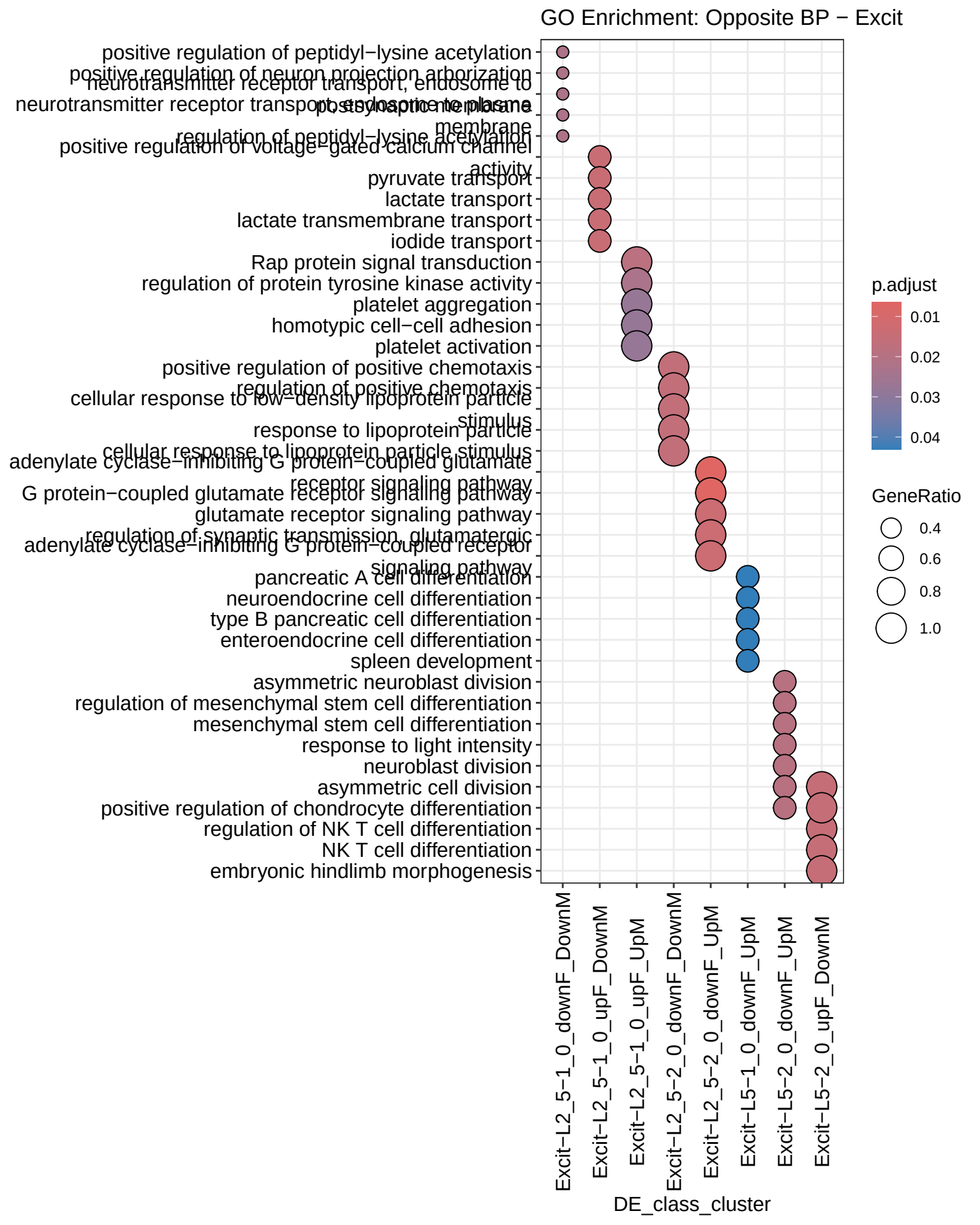

GO Enrichment: Opposite MF − Oligo
cell adhesion mediator activity
extracellular matrix structural constituent conferring tensile strength
collagen binding
active transmembrane transporter activity
inorganic anion transmembrane transporter activity
nucleoside triphosphate diphosphatase activity
cyclosporin A binding
RNA polymerase II CTD heptapeptide repeat peptidyl−prolyl isomerase activity
RNA polymerase II CTD heptapeptide repeat P3 isomerase activity
RNA polymerase II CTD heptapeptide repeat P6 isomerase activity
SNARE binding
glutamate receptor binding
voltage−gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential
calcium−dependent phospholipid binding
metal ion transmembrane transporter activity
sodium channel inhibitor activity
phosphatidic acid binding
phosphatidylinositol−3,4−bisphosphate binding
phosphatidylinositol−3,5−bisphosphate binding
microfilament motor activity
alpha−adrenergic receptor activity
adrenergic receptor activity
neurotrophin binding
structural molecule activity conferring elasticity
S100 protein binding
Oligo−3_0_downF_UpM
Oligo−3_0_upF_DownM
Oligo−3_0_upF_UpM
Oligo−5_0_downF_DownM
Oligo−5_0_upF_UpM
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.1
0.2
0.3
0.4
0.5

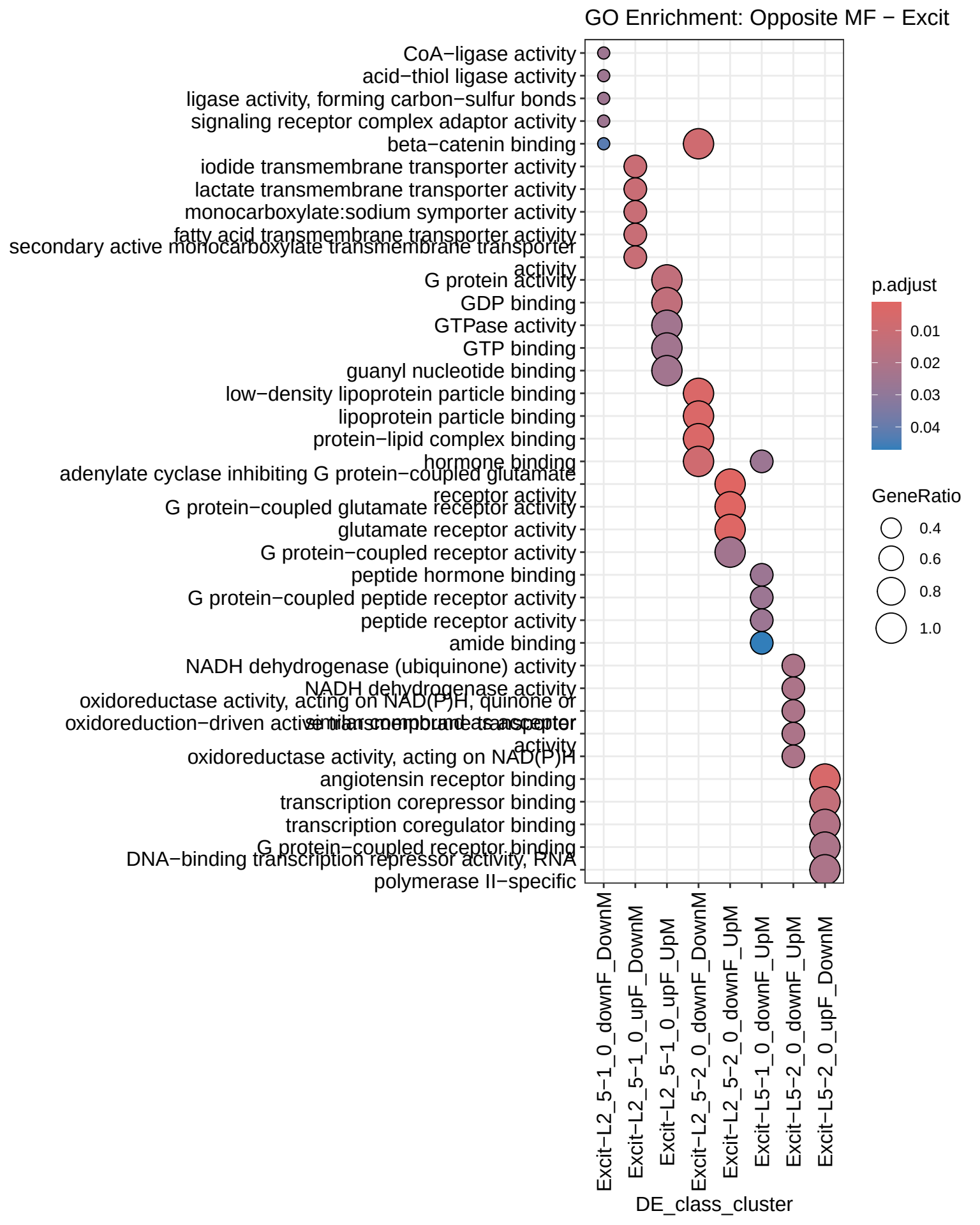

GO Enrichment: Opposite CC − Vasc
phospholipid−translocating ATPase complex
apical plasma membrane
platelet alpha granule membrane
presynaptic active zone cytoplasmic component
apical part of cell
stereocilium base
actin cytoskeleton
chloride channel complex
stereocilium
stereocilium bundle
cytoplasmic dynein complex
dendritic shaft
dynein complex
microtubule associated complex
kinetochore
bicellular tight junction
tight junction
extracellular matrix
external encapsulating structure
interphotoreceptor matrix
Vasc−Endo_0_downF_UpM
Vasc−Endo_0_upF_DownM
Vasc−Endo_0_upF_UpM
Vasc−VLMC_0_downF_UpM
DE_class_cluster
p.adjust
0.03
0.04
GeneRatio
0.1
0.2
0.3
0.4
0.5
0.6

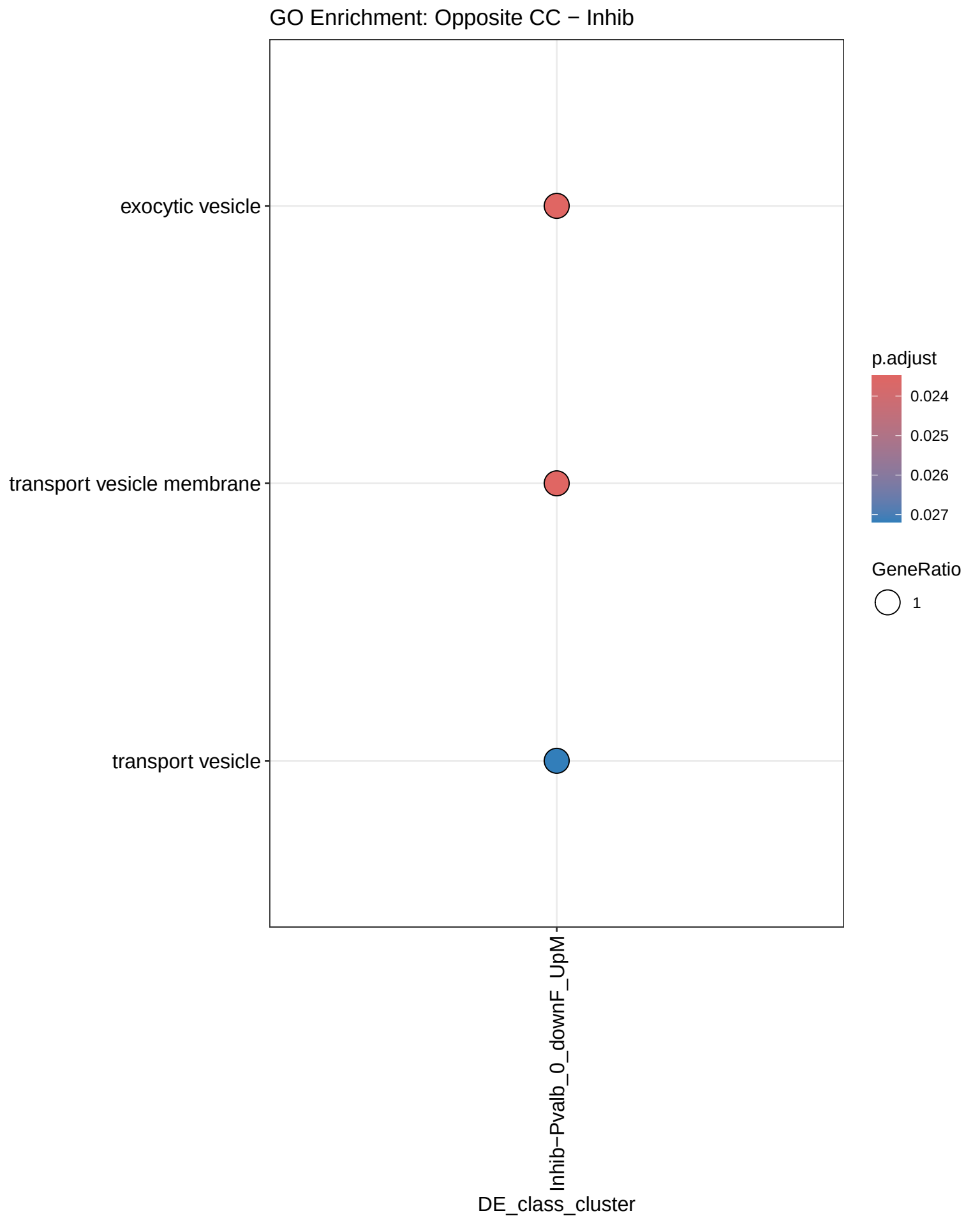

GO Enrichment: Opposite BP − Vasc
aggressive behavior
positive regulation of chromatin binding
glomerulus morphogenesis
positive regulation of receptor catabolic process
regulation of protein localization to early endosome
aminophospholipid translocation
regulation of relaxation of cardiac muscle
regulation of chloride transport
regulation of microvillus assembly
aminophospholipid transport
protein histidyl modification to diphthamide
peptidyl−histidine modification
protein maturation
regulation of membrane protein ectodomain proteolysis
positive regulation of ATP−dependent activity
renal protein absorption
cell−cell junction maintenance
glomerular filtration
renal filtration
renal absorption
Vasc−Endo_0_downF_DownM
Vasc−Endo_0_downF_UpM
Vasc−Endo_0_upF_DownM
Vasc−Endo_0_upF_UpM
DE_class_cluster
p.adjust
0.027
0.030
0.033
0.036
0.039
GeneRatio
0.3
0.4
0.5
0.6

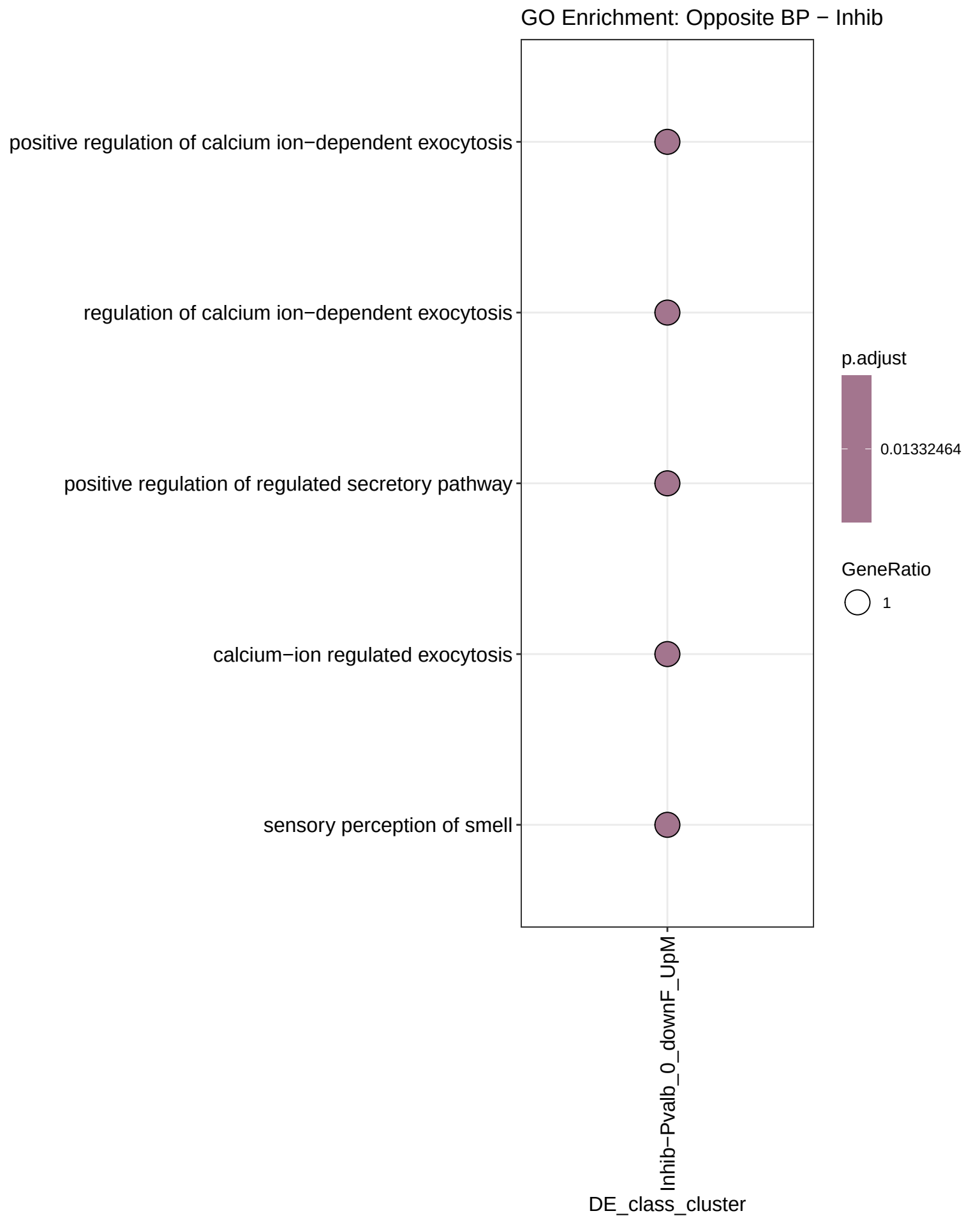
GO Enrichment: Opposite MF − Vasc
calcium, potassium:sodium antiporter activity
calcium:sodium antiporter activity
STAT family protein binding
calcium:monoatomic cation antiporter activity
histone ubiquitin ligase activity
aminophospholipid flippase activity
phosphatidylcholine flippase activity
beta−2 adrenergic receptor binding
phosphatidylcholine floppase activity
phosphatidylserine floppase activity
ATPase activator activity
ferrous iron binding
ATPase regulator activity
chloride channel activity
monoatomic anion channel activity
dynein light chain binding
dynein heavy chain binding
spectrin binding
myosin binding
microtubule motor activity
Roundabout binding
neuroligin family protein binding
integrin binding
ciliary neurotrophic factor receptor binding
laminin−1 binding
Vasc−Endo_0_downF_DownM
Vasc−Endo_0_downF_UpM
Vasc−Endo_0_upF_DownM
Vasc−Endo_0_upF_UpM
Vasc−VLMC_0_downF_UpM
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.2
0.3
0.4
0.5

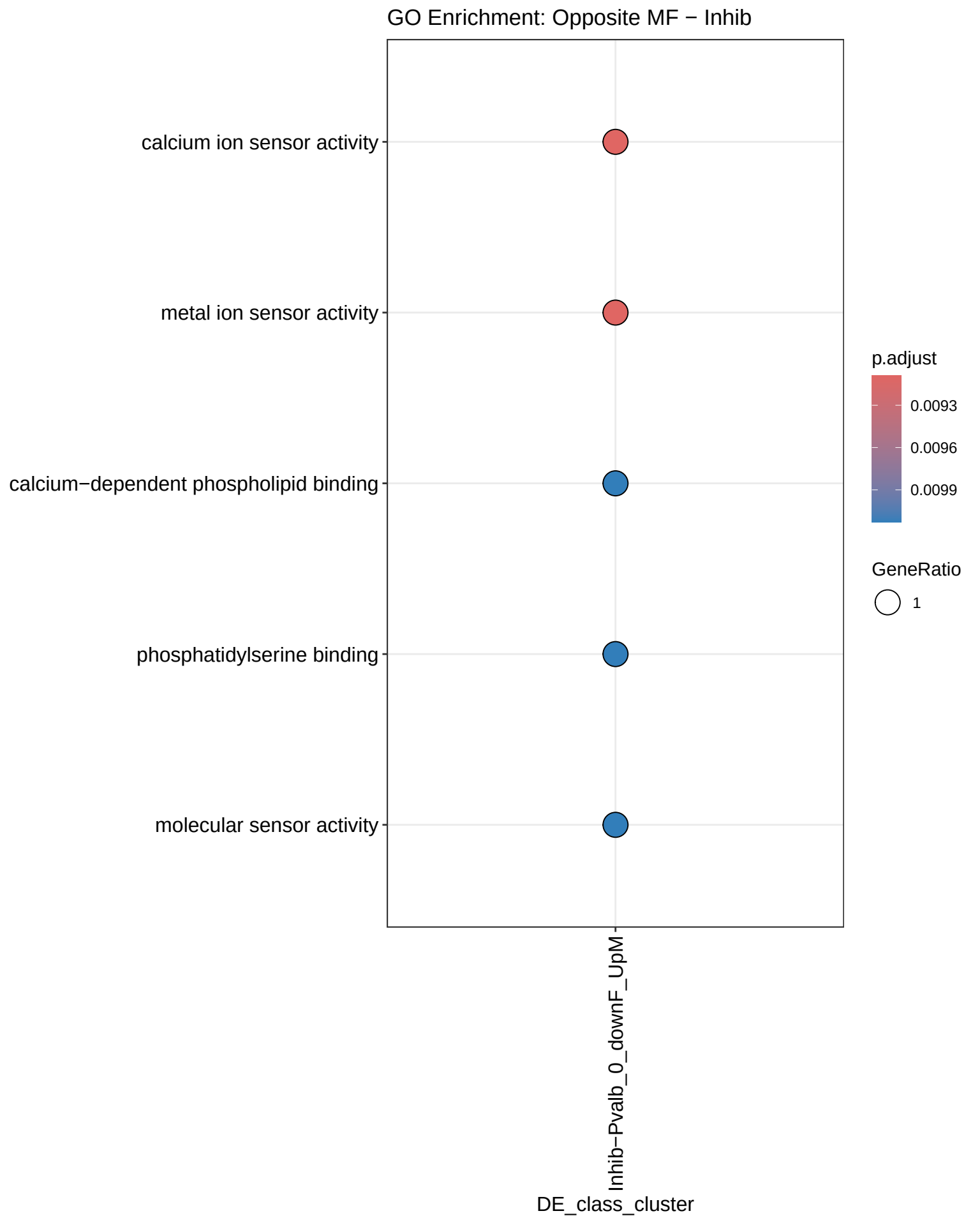

GO Enrichment: Opposite CC − Excit
GINS complex
DNA replication preinitiation complex
sarcolemma
cytoplasmic side of plasma membrane
protein−DNA complex
tertiary granule membrane
cell−cell contact zone
specific granule membrane
recycling endosome membrane
bicellular tight junction
catenin complex
caveola
GABA−ergic synapse
extrinsic component of plasma membrane
plasma membrane raft
NADH dehydrogenase complex
respiratory chain complex I
respiratory chain complex
oxidoreductase complex
male germ cell nucleus
germ cell nucleus
transcription repressor complex
PML body
nuclear speck
Excit−L2_5−1_0_upF_DownM
Excit−L2_5−1_0_upF_UpM
Excit−L2_5−2_0_downF_DownM
Excit−L5−2_0_downF_UpM
Excit−L5−2_0_upF_DownM
DE_class_cluster
p.adjust
0.01
0.02
0.03
GeneRatio
0.5
0.6
0.7
0.8
0.9
1.0


GO Enrichment: Opposite BP − Excit
positive regulation of peptidyl−lysine acetylation
positive regulation of neuron projection arborization
neurotransmitter receptor transport, endosome to
neurotransmitter receptor transport, endosome to plasma
postsynaptic membrane
regulation of peptidyl−lysine acetylation
positive regulation of voltage−gated calcium channel
activity
pyruvate transport
lactate transport
lactate transmembrane transport
iodide transport
Rap protein signal transduction
regulation of protein tyrosine kinase activity
platelet aggregation
homotypic cell−cell adhesion
platelet activation
positive regulation of positive chemotaxis
regulation of positive chemotaxis
cellular response to low−density lipoprotein particle
stimulus
response to lipoprotein particle
cellular response to lipoprotein particle stimulus
adenylate cyclase−inhibiting G protein−coupled glutamate
receptor signaling pathway
G protein−coupled glutamate receptor signaling pathway
glutamate receptor signaling pathway
regulation of synaptic transmission, glutamatergic
adenylate cyclase−inhibiting G protein−coupled receptor
signaling pathway
pancreatic A cell differentiation
neuroendocrine cell differentiation
type B pancreatic cell differentiation
enteroendocrine cell differentiation
spleen development
asymmetric neuroblast division
regulation of mesenchymal stem cell differentiation
mesenchymal stem cell differentiation
response to light intensity
neuroblast division
asymmetric cell division
positive regulation of chondrocyte differentiation
regulation of NK T cell differentiation
NK T cell differentiation
embryonic hindlimb morphogenesis
Excit−L2_5−1_0_downF_DownM
Excit−L2_5−1_0_upF_DownM
Excit−L2_5−1_0_upF_UpM
Excit−L2_5−2_0_downF_DownM
Excit−L2_5−2_0_downF_UpM
Excit−L5−1_0_downF_UpM
Excit−L5−2_0_downF_UpM
Excit−L5−2_0_upF_DownM
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.4
0.6
0.8
1.0


GO Enrichment: Opposite MF − Excit
CoA−ligase activity
acid−thiol ligase activity
ligase activity, forming carbon−sulfur bonds
signaling receptor complex adaptor activity
beta−catenin binding
iodide transmembrane transporter activity
lactate transmembrane transporter activity
monocarboxylate:sodium symporter activity
fatty acid transmembrane transporter activity
secondary active monocarboxylate transmembrane transporter activity
G protein activity
GDP binding
GTPase activity
GTP binding
guanyl nucleotide binding
low−density lipoprotein particle binding
lipoprotein particle binding
protein−lipid complex binding
hormone binding
adenylate cyclase inhibiting G protein−coupled glutamate receptor activity
G protein−coupled glutamate receptor activity
glutamate receptor activity
G protein−coupled receptor activity
peptide hormone binding
G protein−coupled peptide receptor activity
peptide receptor activity
amide binding
NADH dehydrogenase (ubiquinone) activity
NADH dehydrogenase activity
oxidoreductase activity, acting on NAD(P)H, quinone or similar compound as acceptor
oxidoreduction−driven active transmembrane transporter activity
oxidoreductase activity, acting on NAD(P)H
angiotensin receptor binding
transcription corepressor binding
transcription coregulator binding
G protein−coupled receptor binding
DNA−binding transcription repressor activity, RNA polymerase II−specific
Excit−L2_5−1_0_downF_DownM
Excit−L2_5−1_0_upF_DownM
Excit−L2_5−1_0_upF_UpM
Excit−L2_5−2_0_downF_DownM
Excit−L2_5−2_0_downF_UpM
Excit−L5−1_0_downF_UpM
Excit−L5−2_0_downF_UpM
Excit−L5−2_0_upF_DownM
DE_class_cluster
p.adjust
0.01
0.02
0.03
0.04
GeneRatio
0.4
0.6
0.8
1.0


GO Enrichment: Opposite CC − Inhib
exocytic vesicle
transport vesicle membrane
transport vesicle
Inhib−Pvalb_0_downF_UpM
DE_class_cluster
p.adjust
0.024
0.025
0.026
0.027
GeneRatio
1


GO Enrichment: Opposite BP − Inhib
positive regulation of calcium ion−dependent exocytosis
regulation of calcium ion−dependent exocytosis
positive regulation of regulated secretory pathway
calcium−ion regulated exocytosis
sensory perception of smell
Inhib−Pvalb_0_downF_UpM
DE_class_cluster
p.adjust
0.01332464
GeneRatio
1


GO Enrichment: Opposite MF − Inhib
calcium ion sensor activity
metal ion sensor activity
calcium−dependent phospholipid binding
phosphatidylserine binding
molecular sensor activity
Inhib−Pvalb_0_downF_UpM
DE_class_cluster
p.adjust
0.0093
0.0096
0.0099
GeneRatio
1